CHAPTER L-06

PRICING AND VALUATION IN LIFE INSURANCE

Chapter Introduction

The objective of this chapter is to introduce to the learner the basic elements that are involved in the pricing and benefits of life insurance contracts. We shall first discuss the elements that constitute the premium and then discuss the concept of surplus and bonus.

Learning Outcomes

A. Insurance pricing – basic elements

B. Surplus and bonus

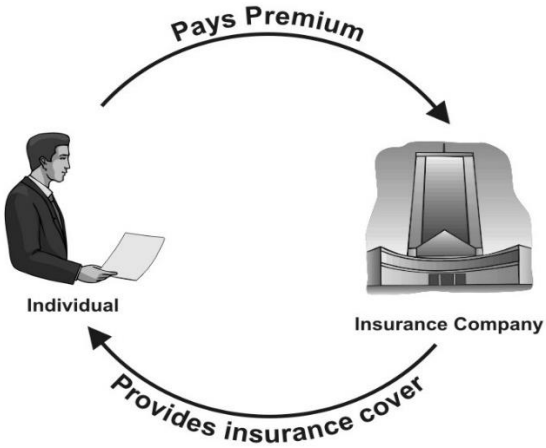
A. Insurance pricing – Basic elements

1. Premium

In ordinary language, the term premium denotes the price that is paid by an insured for purchasing an insurance policy. It is normally expressed as a rate of premium per thousand rupees of sum assured. The premium rates depend on the age of the prospect and the plan.

These premium rates are available in the form of tables of rates that are available with insurance companies.

Diagram 1: Premium



The rates printed in these tables are known as "Office Premiums". They are in most cases the same throughout the term and are expressed as an annual rate.

Example

If the premium for a twenty year endowment policy for a given age is Rs. 4,800, it means that Rs. 4,800 has to be paid each year for twenty years.

However it is possible to have some policies in which the premiums are payable only in the first few years. Companies also have single premium contracts in which only one premium is payable at the beginning of the contract. These policies are usually investment oriented.

2. Rebates

Life insurance companies may also offer certain types of rebates on the premium that is payable. Two such rebates are:

- ✓ For sum assured
- ✓ For mode of premum

Rebate for sum assured

The rebate **for sum assured** is offered to those who buy policies with higher amounts of sum assured. It is offered as a way of passing on to the customer, the gains that the insurer may make when servicing higher value policies. The logic is that the effort and cost required to process a policy of Rs 50,000 or 5,00,000 remains the same. But higher sum assured policies yield more premium and so more profits.

Rebate for mode of premium

Similarly a rebate may be offered **for the mode of premium**. Life insurance companies may allow premiums to be paid on annual, half yearly, quarterly or monthly basis. More frequent the mode, more the administrative costs for collecting and accounting the premium. Again, in the yearly mode, the insurer can utilise this amount during the entire year and earn interest on it. Insurers would hence encourage payment via yearly and half yearly modes by allowing a rebate on these. They may also charge a little extra for monthly mode of payments, to cover additional administrative expenses involved.

3. Extra charges

The tabular premium is charged for those individuals who are not subject to any significant factors that would pose an extra risk. They are known as **standard lives** and the rates charged are known as ordinary rates.

If a person proposing for insurance suffers from certain health problems like heart ailments or diabetes that can pose a hazard to his life, he or she is considered to be sub-standard. The insurer may decide to impose an extra premium by way of a health extra. Similarly an occupational extra may be imposed on those engaged in a hazardous occupation, like a circus acrobat. These extras would result in the premium being more than the tabular premium.

Again, an insurer may offer certain extra benefits under a policy, which are available on payment of an extra premium.

Example

A life insurer may offer a Double Accident Benefit or DAB (where double the sum assured is payable as a claim if death is a result of accident). For this it may charge an extra premium of one rupee per thousand sum assured.

Similarly a benefit known as Permanent Disability Benefit (PDB) may be availed by paying an extra per thousand sum assured.

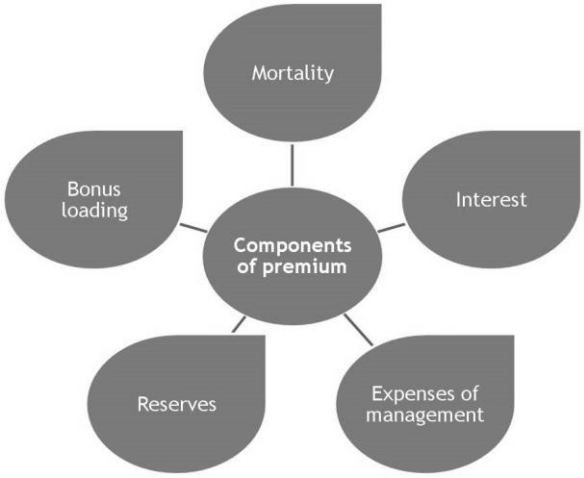
4. Determining the premium

Let us now examine how life insurers arrive at the rates that are presented in the premium tables. This task is performed by an actuary. The process of setting

the premium in case of traditional life insurance policies like term insurance, whole life and endowment considers following elements:

- ✓ Mortality
- ✓ Interest
- ✓ Expenses of management
- ✓ Reserves
- ✓ Bonus loading

Diagram 2: Components of Premium



The first two elements give us the Net premium. By adding [also called 'loading'] the other elements to the net premium we get the gross or office premium

a) Mortality and Interest

Mortality is the first element in premiums. It is the chance or likelihood that a person of a certain age would die during a given year. To find out the expected Mortality of a person, "Mortality Tables" are used.

Example

If the mortality rate for age 35 is 0.0035 it implies that out of every 1000 people who are alive as on age 35, 3.5 (or 35 out of 10,000) are expected to die between age 35 and 36.

The table may be used to calculate mortality cost for different ages. For example the rate of 0.0035 for age 35 implies a cost of insurance of 0.0035 x 1000 (sum assured) = Rs. 3.50 per thousand sum assured.

The above cost may be also called the "Risk Premium". For higher ages the risk premium would be higher.

Example

If we need to have Rs. 5 per thousand to meet the cost of insurance after five years and if we assume a rate of interest of 6%, the present value of Rs. 5 payable after five years would be $5 \times 1/ (1.06)^5 = 3.74$.

If instead of 6% we were to assume 10%, the present value would be only 3.10. In other words the higher the rate of interest assumed, the lower the present value.

From our study of mortality and interest there are two major conclusions we can derive

- ✓ Higher the mortality rate in the mortality table, higher the premiums would be
- ✓ Higher the interest rate assumed, lower the premium

Net premium
The estimates of mortality and interest give the "Net Premium"

Gross premium
Gross premium is the net premium plus an amount called loading. There are three considerations or guiding principles that needs to be borne in mind when determining the amount of loading:

b) Expenses and reserves

Life insurers have to incur various types of operating expenses including:

- ✓ Agents training and recruitment,
- ✓ Commissions of agents,
- ✓ Staff salaries,
- ✓ Office accommodation,
- ✓ Office stationery,
- ✓ Electricity charges,
- ✓ Other miscellaneous etc.

All these have to be paid from premiums that are collected by insurers. These expenses are suitably loaded to the net premium.

c) Lapses and contingencies

In addition to expenses, there are other factors that can make the calculations of life insurers go wrong.

One source of risk is that of lapses and withdrawals. A lapse means that the policyholder discontinues payment of premiums. In case of withdrawals, the policyholder surrenders the policy and receives an amount from the policy's acquired cash value.

Lapses usually happen within the first three years, especially in the first year of the contract.

d) With Profit (participating) policies and Bonus loading

The concept of 'With Profit' policies originated when Life insurers started the practice of charging a high loading in advance to create a buffer to keep them solvent even in adverse situations. If subsequent experience proved to be more favourable, the life insurer would share some of the profits it made as a result with policy holders by way of bonus.

In sum we can say that:

Gross premium = Net premium + Loading for expenses + Loading for contingencies + Bonus loading

Test Yourself 1

What does a policy lapse mean?

I. Policyholder completes premium payment for a policy
II. Policyholder discontinues premium payment for a policy
III. Policy attains maturity
IV. Policy is withdrawn from the market

B. Surplus and bonus

1. Determination of surplus and bonus

Every life insurance company is expected to undertake a periodic valuation of its assets and liabilities. Such a valuation has two purposes:

i. To assess the financial state of the life insurer and determine if it is solvent or insolvent
ii. To determine the surplus available for distribution among policyholders/ share holders

Definition

Surplus is the excess of value of assets over value of liabilities. If it is negative, it is known as a strain.

Let us now see how the concept of surplus in life insurance is different from that of profit of a firm.

Firms in general look at profits in two ways. Firstly, profit is the **excess of income over outgo** for a given accounting period, as it appears in the profit and loss account. Profit also forms part of the balance sheet of a firm - it may be defined as the **excess of assets over liabilities**. In both instances, profits are determined at the end of the accounting period.

Surplus = Assets - Liabilities

Let us understand what liabilities mean in life insurance. For a given block of life insurance policies, the life insurer has to make provision for meeting future claims, expenses and other expected pay-outs that may arise. The insurer also expects to receive premiums in future for these policies.

Liabilities are thus the present value of all payments that have to be made less the present value of premiums expected to be received on these policies. The present value is arrived at by applying a suitable rate of discount [the interest rate]
Surplus arises as a result of the life insurer's actual experience being better than what it had assumed. Life insurers are obliged to share the benefits arising as a result with holders of it's with profit policies.

Example

The profits of XYZ firm as on 31st March 2013, is given as its income less expenses or its assets less liabilities as on that date.

In both instances, the profit is clearly defined and is known.

2. Bonus

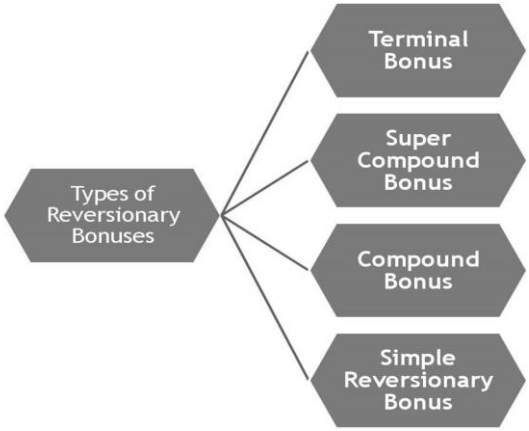
Insurers have to declare and distribute its divisible surplus among the policy holders and shareholders of the company [if any] in the form of a bonus. In India, the United Kingdom and many other countries, distribution of surplus is popular.

Bonus is paid as an addition to the basic benefit payable under a contract. Typically it may appear as an addition to basic sum assured or basic pension per annum. It is expressed, for example, as Rs. 60 per thousand sum assured

The most common form of bonus is the **reversionary bonus**. Once declared these bonus additions, made each year, get attached to the policy and cannot be taken away. They are called 'Reversionary' bonuses because they are received only at the time of a claim by death or maturity. Bonuses may also be payable on surrender provided the contract is eligible through having run for a minimum term [say 5 years]

Types of reversionary bonuses

Diagram 3: Types of Reversionary Bonuses



i. Simple Reversionary Bonus

This is a bonus expressed as a percentage of the basic cash benefit under the contract. In India for example, it is declared as amount per thousand sum assured.

ii. Compound Bonus

Here the company expresses a bonus as a percentage of basic benefit and already attached bonuses. It is thus a bonus on a bonus. A way to express it may be as @ 8% of basic sum assured plus attached bonus.

iii. Terminal Bonus

As the name suggests, this bonus attaches to the contract only at the time of its termination [by death or maturity]. It is applicable only for the claims arising in the ensuing year. Thus terminal bonus declared for 2013 would only apply to claims that have arisen during 2013-14 and not for subsequent years. Terminal bonuses depend on the time duration of the contract and increase with it. A contract that has run for 25 years would have higher terminal bonus than one which has run for 15 years.

3. The Contribution Method

Another method of distribution of surplus adopted in North America is the "Contribution" method. Here, the surplus, i.e. the difference between what was expected to happen and what actually happened over the year with respect to mortality, interest and expenses is declared and distributed as dividends.

The dividends can be paid in cash, by way of adjustments/ reductions in future premiums, by allowing purchase of non-forfeitable paid up additions to the policy or as accumulations to the credit of the policy.

4. Unit Linked Policies

The Principles of Pricing and other features of ULIP Policies have already been covered in an earlier chapter.

Summary

- In ordinary language, the term premium denotes the price that is paid by an insured for purchasing an insurance policy.
- The process of setting the premium for life insurance policies involves consideration of mortality, interests, expense management and reserves.
- Gross premium is the net premium plus an amount called loading.
- A lapse means that the policyholder discontinues payment of premiums. In case of withdrawals, the policyholder surrenders the policy and receives an amount from the policy's acquired cash value.
- Surplus arises as a result of the life insurer's actual experience being better than what it had assumed.
- Surplus allocation could be towards maintaining solvency requirements, increasing free assets etc.
- The most common form of bonus is the reversionary bonus.

Key Terms

1. Premium
2. Rebate
3. Bonus
4. Surplus
5. Reserve
6. Loading
7. Reversionary bonus

Answers to Test Yourself

Answer 1 - The correct option is II.